HYOSCINE BUTYLBROMIDE

Not hyoscine hydrobromide

No CNS effects

CHEMICAL NAME (1S,2S,4R,5R,7s)-9-butyl-7-[(2S)-3-hydroxy-2-phenylpropanoyl]oxy-9-methyl-3-oxa-9-azoniatricyclo[3.3.1.0²,⁴]nonane bromide

CLASS Antimuscarinic antispasmodic (quaternary — peripheral)

TRADE NAMES Buscopan · Scopinal · Spasmonal

THERAPEUTIC USES / INDICATIONS

- GI and genitourinary smooth muscle spasm: abdominal cramps, IBS, biliary and renal colic.
- Dysmenorrhoea.
- Reduce bowel motility for radiology/endoscopy.
- Palliative care: excessive respiratory secretions (death rattle), bowel colic.

ADMINISTRATION

- **PO** tablets.
- **IV** slowly or **IM**; **SC** (palliative).
- Do not confuse with hyoscine hydrobromide (crosses BBB; different dosing).

DOSAGES

- **PO:** 20 mg four times daily (IBS: 10 mg TDS, up to 20 mg QDS).
- **IV/IM (acute spasm):** 20 mg, repeat after 30 min; max 100 mg/day.
- **Palliative secretions:** 20 mg SC, then 60–120 mg/24 h by syringe driver.
- **Children 6–11 yr:** 10 mg PO three times daily.

PHARMACOKINETICS

- **Onset:** IV 3–5 min; PO 1–2 h (poor absorption ~1%). **Duration:** 2–6 h.
- Quaternary — does not cross blood–brain barrier.
- Excreted in urine and faeces. **t½:** 5 h.

PHARMACODYNAMICS

- Blocks **muscarinic receptors** on smooth muscle → relaxes GI, biliary, urinary and uterine spasm; ↓ secretions.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry mouth, tachycardia.
- Blurred vision (accommodation).
- Constipation, urinary hesitancy.

ADVERSE EFFECTS (SERIOUS)

- Serious cardiac adverse effects (tachycardia, hypotension, cardiac arrest) with IV/IM in patients with heart disease.
- Urinary retention.
- Angle-closure glaucoma; anaphylaxis.

CONTRAINDICATIONS & CAUTIONS

- Myasthenia gravis; megacolon; mechanical GI obstruction; paralytic ileus.
- Untreated angle-closure glaucoma.
- Tachycardia (IV/IM); prostatic hypertrophy with retention.

KEY INTERACTIONS

- **Other anticholinergics (TCAs, antihistamines):** additive.
- **Metoclopramide, domperidone:** mutual antagonism.

NURSING MANAGEMENT

- IV/IM: monitor HR, BP; caution in cardiac disease.
- Check for abdominal distension, bowel sounds.
- Monitor urine output.
- Palliative: assess secretions and comfort.

PATIENT EDUCATION

- May cause dry mouth and blurred vision; avoid driving if affected.
- Report eye pain, difficulty passing urine, fast heartbeat.
- See a doctor if abdominal pain persists or with fever, vomiting or blood in stools.

KEY POINTS

- Peripheral antimuscarinic antispasmodic.
- No CNS effects (quaternary).
- 20 mg QDS PO; 20 mg IV/IM.
- Caution IV in heart disease; not hydrobromide.